An Exact Modulo-Five Obstruction for OEIS A128925

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Abstract

OEIS A128925 consists of primes p for which at least one of $p^2 - 6$ and $p^2 + 6$ is prime. Its entry states that 5 is the unique term for which both are prime. We give a complete elementary proof and identify, by residue class, which of the two expressions must be composite. This verifies an existing OEIS comment, not a new open conjecture.

OEIS assertion and precise theorem

The OEIS entry A128925, offset 1, lists primes p for which $p^2 - 6$ or $p^2 + 6$ is prime and comments that $p = 5$ is the only member for which both are prime [1].

Theorem 1. *For any prime p, both $p^2 - 6$ and $p^2 + 6$ are prime if and only if $p = 5$. More precisely, for $p > 5$:*

1. *if $p \equiv \pm 1 \pmod 5$, then $5 \mid p^2 - 6$ and $p^2 - 6 > 5$;*
2. *if $p \equiv \pm 2 \pmod 5$, then $5 \mid p^2 + 6$ and $p^2 + 6 > 5$.*

Thus, for every prime $p > 5$, at most one of these two expressions can be prime.

Proof. Every prime $p > 5$ is nonzero modulo 5. Its residue is ± 1 or ± 2, hence $p^2 \equiv 1$ or $4 \pmod 5$. If $p^2 \equiv 1$, then $p^2 - 6 \equiv 0 \pmod 5$. Since $p \geq 7$, $p^2 - 6 \geq 43 > 5$, so this expression is composite. If $p^2 \equiv 4$, then $p^2 + 6 \equiv 0 \pmod 5$ and $p^2 + 6 \geq 55 > 5$, so this expression is composite.

For the remaining primes, $p = 2$ gives $p^2 - 6 = -2$ and $p^2 + 6 = 10$, neither prime. For $p = 3$, the two values are 3 and 15, so they are not both prime. Finally, for $p = 5$, the two values are 19 and 31, both prime. This proves the assertion. □

Scope and checks

The result settles only the OEIS comment concerning simultaneous primality; it does *not* determine whether A128925 is infinite, or prove that either sign is prime infinitely often. The argument is an elementary modular obstruction, so no novelty or priority claim is made. A direct check for $p = 2, 3, 5, 7, 11, 13, 17$ agrees with the theorem. The two sign classes partition all primes $p > 5$.

References

1. J. M. Bergot et al., "A128925: Primes p such that at least one of $p^2 - 6$, $p^2 + 6$ is prime," *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org/A128925`, consulted November 14, 2026.

AI disclosure. This note was generated and mathematically checked by ChatGPT (OpenAI). It has not undergone independent human peer review.